Equations Used — Economic Fibrillation Research

Every formula in the pipeline, from raw GDP to the significance test. For each: the equation, its symbols, and one line on what it does here.

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1. Natural-log transform

$$g_i(t) = \ln G_i(t)$$

- $G_i(t)$ — GDP of economy i in year t
- $g_i(t)$ — its natural logarithm

Use here: Turns exponential GDP growth into a near-straight line, so the business cycle becomes a simple wobble on top.

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2. Cyclical component (band-pass detrending)

$$x_i(t) = \mathcal{B}[\, g_i(t) - \bar{g}_i \,]$$

- $\mathcal{B}[\cdot]$ — Butterworth band-pass filter (keeps ≈2–9 year cycles)
- $\bar{g}_i$ — time-average of g_i
- $x_i(t)$ — the business-cycle wobble

Use here: Strips away the trend and keeps only the business cycle — the part that actually oscillates.

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3. Hilbert transform

$$H[x](t) = \frac{1}{\pi}\,\mathrm{P.V.}\int_{-\infty}^{\infty} \frac{x(\tau)}{t-\tau}\,d\tau$$

- $x(\tau)$ — the cyclical signal
- P.V. — Cauchy principal value of the integral
- $H[x](t)$ — the 90°-shifted partner signal

Use here: Creates the quadrature partner of the wobble, so each economy can be given a rotating angle.

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4. Analytic signal

$$z_i(t) = x_i(t) + i\,H[x_i](t) = A_i(t)\,e^{\,i\theta_i(t)}$$

- $z_i(t)$ — complex analytic signal
- $A_i(t)$ — amplitude
- $\theta_i(t)$ — phase
- i — imaginary unit

Use here: Packs the wobble and its partner into one rotating arrow (a clock hand) for each economy.

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5. Instantaneous amplitude

$$A_i(t) = \sqrt{x_i(t)^2 + H[x_i](t)^2}$$

- $A_i(t)$ — size of the boom/bust swing at time t

Use here: Measures how strong an economy's cycle is at each moment.

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6. Instantaneous phase

$$\theta_i(t) = \text{atan2}(H[x_i](t),\, x_i(t))$$

- $\theta_i(t)$ — clock-hand angle (position in the cycle)
- atan2 — angle of the point (x, H[x])

Use here: Tells us exactly where each economy sits in its boom–bust cycle — the core quantity of the whole study.

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7. Instantaneous frequency

$$\omega_i(t) = \frac{d\theta_i}{dt}$$

- $\omega_i(t)$ — how fast economy i moves through its cycle

Use here: Measures the local speed of each economy's cycle.

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8. Kuramoto order parameter

$$r(t)\, e^{\,i\psi(t)} = \frac{1}{N} \sum_{j=1}^{N} e^{\,i\theta_j(t)}$$

- r(t) — synchronization (0 = scattered, 1 = locked)
- ψ(t) — average phase
- N — number of economies (26)

Use here: The single number we track: how synchronized all the economies are in a given year.

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9. Kuramoto model (dynamics)

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^{N} \sin(\theta_j - \theta_i) = \omega_i + K\, r \sin(\psi - \theta_i)$$

- K — coupling strength (how hard economies pull each other)
- ω_i — natural frequency
- right form — the mean-field version using r and ψ

Use here: The rule for how coupled economies pull each other into sync; the basis for the fibrillation mechanism.

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10. Critical coupling

$$K_c = \frac{2}{\pi g(0)}$$

- K_c — coupling threshold where sync switches on
- g(0) — peak of the spread of natural frequencies

Use here: The tipping point: above K_c economies lock together; dropping below it is 'coupling collapse' — fibrillation.

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11. Networked Kuramoto (connection map)

$$\frac{d\theta_i}{dt} = \omega_i + \sum_{j=1}^{N} K_{ij} \sin(\theta_j - \theta_i)$$

- K_ij — strength of the link from economy j to economy i (the connection map)

Use here: Upgrades the model so a shock can spread along real connections from an epicenter — the planned next stage.

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12. Circular-shift surrogate

$$x_i^{(s)}(t) = x_i((t + \tau_i) \bmod T)$$

- τ_i — random time-shift for economy i
- T — length of the series
- $x_i^{(s)}$ — one surrogate ('fake') series

Use here: Builds a fake world that keeps each economy's own rhythm but destroys the alignment — the ruler for the test.

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13. Fourier surrogate

$$x_i^{(s)} = \mathcal{F}^{-1}\left\{ |X_i(f)|\, e^{i\xi(f)} \right\}, \quad \xi(f) \sim U(0, 2\pi)$$

- $|X_i(f)|$ — amplitude spectrum (kept)
- ξ(f) — random phases
- $\mathcal{F}^{-1}$ — inverse Fourier transform

Use here: A second kind of fake economy that keeps the power spectrum but randomizes timing, for robustness.

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14. Relative-test contrast

$$\Delta = \bar{r}_{\text{crisis}} - \bar{r}_{\text{rest}}$$

- r̄_crisis — mean synchronization in global-crisis years
- r̄_rest — mean in all other years

- Δ — the gap between them

Use here: Measures how much more synchronized the world is during global crises than the rest of the time.

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15. Monte Carlo p-value

$$p = \frac{1 + \#\{\,\Delta^{(s)} \geq \Delta\,\}}{1 + M}$$

- Δ — the real contrast
- $\Delta^{(s)}$ — contrast from surrogate s
- M — number of surrogates
- #{·} — count of

Use here: The chance the crisis synchronization is just luck; a small p means the result is real.
